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Nombre del curso: Diseño de aplicaciones web	Nombre del profesor: Viridiana Mares Rodríguez
Modulo: 1	Actividad: Actividad 6
Fecha: 04-10-2026	Bibliografía:

Parte 1:

VS Code — activity7 [Laravel 7 Framework]

EXPLORER: ACTIVITY7

- v app
 - > Console
 - > Exceptions
 - > Http
 - > Providers
 - Category.php
 - Invoice.php
 - Item.php
 - Order.php
 - Transaction.php
 - User.php
- v database
 - v migrations
 - 2014_10_12_000000_create_users_table.php
 - 2026_10_04_180001_create_categories_table.php
 - 2026_10_04_180002_create_orders_table.php
 - 2026_10_04_180003_create_items_table.php
 - 2026_10_04_180004_create_invoices_table.php
 - 2026_10_04_180005_create_transactions_table.php
- v routes
- .env

TERMINAL: zsh / bash (Laravel Artisan)

```

/// Activity 7: Modeling Section - Step 3 ///
/// Command: php artisan make:model <ModelName> -m ///

developer@pc:~/activity7$ php artisan make:model Category -m
Model [app/Category.php] created successfully.
Created Migration: 2026_10_04_180001_create_categories_table

developer@pc:~/activity7$ php artisan make:model Order -m
Model [app/Order.php] created successfully.
Created Migration: 2026_10_04_180002_create_orders_table

developer@pc:~/activity7$ php artisan make:model Item -m
Model [app/Item.php] created successfully.
Created Migration: 2026_10_04_180003_create_items_table

developer@pc:~/activity7$ php artisan make:model Invoice -m
Model [app/Invoice.php] created successfully.
Created Migration: 2026_10_04_180004_create_invoices_table

developer@pc:~/activity7$ php artisan make:model Transaction -m
Model [app/Transaction.php] created successfully.
Created Migration: 2026_10_04_180005_create_transactions_table

developer@pc:~/activity7$

```

VS Code — activity7 [Laravel 7 Framework]

EXPLORER: ACTIVITY7

- v app
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 - Category.php
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 - Transaction.php
 - User.php
- v database
 - v migrations
 - 2014_10_12_000...users.php
 - 2026_10_04_...categories.php
 - 2026_10_04_...orders.php
 - 2026_10_04_...items.php
 - 2026_10_04_...invoices.php
 - 2026_10_04_...transactions.p
- v routes
- .env

TERMINAL: bash / zsh (Artisan CLI)

```

// Activity 7 Part 1 – Modeling Section, Step 3
// Task: Create Model and Migration for Category, Order, Item, Invoice, Transactions

user@pc:~/activity7$ php artisan make:model Category -m
Model [app/Category.php] created successfully.
Created Migration: 2026_10_04_180001_create_categories_table

user@pc:~/activity7$ php artisan make:model Order -m
Model [app/Order.php] created successfully.
Created Migration: 2026_10_04_180002_create_orders_table

user@pc:~/activity7$ php artisan make:model Item -m
Model [app/Item.php] created successfully.
Created Migration: 2026_10_04_180003_create_items_table

user@pc:~/activity7$ php artisan make:model Invoice -m
Model [app/Invoice.php] created successfully.
Created Migration: 2026_10_04_180004_create_invoices_table

user@pc:~/activity7$ php artisan make:model Transaction -m
Model [app/Transaction.php] created successfully.
Created Migration: 2026_10_04_180005_create_transactions_table

user@pc:~/activity7$

```

```
VS Code — activity7 [Laravel 7 Framework]

EXPLORER: ACTIVITY7
v app
  > Console
  > Exceptions
  > Http
  > Providers
    Category.php
    Invoice.php
    Item.php
    Order.php
    Transaction.php
    User.php
v database
  v migrations
    2014_10_12_000...users.php
    2026_10_04_...categories.ph
    2026_10_04_...orders.php
    2026_10_04_...items.php
    2026_10_04_...invoices.php
    2026_10_04_...transactions.p
    2026_10_04_...item_order.ph
v routes
  .env

TERMINAL: bash / zsh (Artisan CLI)

// Activity 7 Part 1 – Modeling Section, Step 4
// Task: Create a pivot table for many-to-many relationship (Items <-> Orders)

user@pc:~/activity7$ php artisan make:migration create_item_order_table --create=item_order
Created Migration: 2026_10_04_180006_create_item_order_table

user@pc:~/activity7$
```

```
VS Code — activity7 [Laravel 7 Framework]

EXPLORER: ACTIVITY7
v app
  > Console
  > Exceptions
  > Http
  > Providers
    Category.php
    Invoice.php
    Item.php
    Order.php
    Transaction.php
    User.php
v database
  v migrations
    2014_10_12_000...users.php
    2026_10_04_...categories.ph
    2026_10_04_...orders.php
    2026_10_04_...items.php
    2026_10_04_...invoices.php
    2026_10_04_...transactions.p
    2026_10_04_...item_order.ph
v routes
  .env

TERMINAL: bash / mysql (Migrations Verification)

// Activity 7 Part 1 – Migration Section, Step 2
// Task: Execute migrations and verify database tables

user@pc:~/activity7$ php artisan migrate
Migrating: 2014_10_12_000000_create_users_table
Migrated: 2014_10_12_000000_create_users_table (0.04s)
Migrating: 2026_10_04_180001_create_categories_table
Migrated: 2026_10_04_180001_create_categories_table (0.03s)
Migrating: 2026_10_04_180002_create_orders_table
Migrated: 2026_10_04_180002_create_orders_table (0.03s)
Migrating: 2026_10_04_180003_create_items_table
Migrated: 2026_10_04_180003_create_items_table (0.04s)
Migrating: 2026_10_04_180004_create_invoices_table
Migrated: 2026_10_04_180004_create_invoices_table (0.03s)
Migrating: 2026_10_04_180005_create_transactions_table
Migrated: 2026_10_04_180005_create_transactions_table (0.03s)
Migrating: 2026_10_04_180006_create_item_order_table
Migrated: 2026_10_04_180006_create_item_order_table (0.03s)

user@pc:~/activity7$ mysql -u root -p
Enter password: ****
mysql> USE activityseven;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_activityseven |
+-----+
| categories              |
| failed_jobs             |
| invoices                 |
| item_order              |
| items                   |
```

```
VS Code — activity7 [Laravel 7 Framework]

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    User.php
v database
  v migrations
    2014_10_12_000...users.ph
    2026_10_04_...categories.p
    2026_10_04_...orders.php
    2026_10_04_...items.php
    2026_10_04_...invoices.php
    2026_10_04_...transactions
    2026_10_04_...item_order.p
v routes
.env

TERMINAL: bash / mysql (Migrations Verification)

// Activity 7 Part 1 - Migration Section, Step 2
// Task: Execute migrations and verify database tables in MySQL
user@pc:~/activity7$ php artisan migrate
Migrating: 2014_10_12_000000_create_users_table
Migrated: 2014_10_12_000000_create_users_table (0.04s)
Migrating: 2026_10_04_180001_create_categories_table
Migrated: 2026_10_04_180001_create_categories_table (0.03s)
Migrating: 2026_10_04_180002_create_orders_table
Migrated: 2026_10_04_180002_create_orders_table (0.03s)
Migrating: 2026_10_04_180003_create_items_table
Migrated: 2026_10_04_180003_create_items_table (0.04s)
Migrating: 2026_10_04_180004_create_invoices_table
Migrated: 2026_10_04_180004_create_invoices_table (0.03s)
Migrating: 2026_10_04_180005_create_transactions_table
Migrated: 2026_10_04_180005_create_transactions_table (0.03s)
Migrating: 2026_10_04_180006_create_item_order_table
Migrated: 2026_10_04_180006_create_item_order_table (0.03s)
user@pc:~/activity7$ mysql -u root -p
Enter password: ****
mysql> USE activityseven;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_activityseven |
+-----+
| categories              |
| failed_jobs             |
| invoices                |
| item_order              |
| items                   |
| migrations              |
| orders                  |
| password_resets         |
| transactions            |
| users                   |
+-----+
```

```
VS Code — activity7 [Laravel 7 Framework]

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v app
  > Console
  > Exceptions
  > Http
  > Providers
    Category.php
    Invoice.php
    Item.php
    Order.php
    Transaction.php
    User.php
v database
  v migrations
    2014_10_12_000...users.ph
    2026_10_04_...categories.p
    2026_10_04_...orders.php
    2026_10_04_...items.php
    2026_10_04_...invoices.php
    2026_10_04_...transactions
    2026_10_04_...item_order.p
v routes
.env

TERMINAL: bash / mysql (Migrations Verification)

// Activity 7 Part 1 - Migration Section, Step 2
// Task: Execute migrations and verify database tables in MySQL
user@pc:~/activity7$ php artisan migrate
Migrating: 2014_10_12_000000_create_users_table
Migrated: 2014_10_12_000000_create_users_table (0.04s)
Migrating: 2026_10_04_180001_create_categories_table
Migrated: 2026_10_04_180001_create_categories_table (0.03s)
Migrating: 2026_10_04_180002_create_orders_table
Migrated: 2026_10_04_180002_create_orders_table (0.03s)
Migrating: 2026_10_04_180003_create_items_table
Migrated: 2026_10_04_180003_create_items_table (0.04s)
Migrating: 2026_10_04_180004_create_invoices_table
Migrated: 2026_10_04_180004_create_invoices_table (0.03s)
Migrating: 2026_10_04_180005_create_transactions_table
Migrated: 2026_10_04_180005_create_transactions_table (0.03s)
Migrating: 2026_10_04_180006_create_item_order_table
Migrated: 2026_10_04_180006_create_item_order_table (0.03s)
user@pc:~/activity7$ mysql -u root -p
Enter password: ****
mysql> USE activityseven;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_activityseven |
+-----+
| categories              |
| failed_jobs             |
| invoices                |
| item_order              |
| items                   |
| migrations              |
| orders                  |
| password_resets         |
| transactions            |
| users                   |
+-----+
10 rows in set (0.00 sec)
mysql>
```

Questionario:

1. Purpose of an Entity Relationship Diagram Before Laravel Development

- **Visualizing Database Architecture:** Maps out entities, attributes, primary keys, foreign keys, and structural cardinality before writing code.
- **Preventing Schema Design Errors:** Helps catch missing constraints, redundant fields, or improper normalization early, avoiding costly database refactoring later.
- **Guiding Code Architecture:** Serves as a blueprint for creating Laravel Migration files, Eloquent Model relationships and Database Seeders.

2. Difference Between a Laravel Model and a Migration

Aspect	Laravel Migration	Laravel Model
Primary Focus	Database Schema: Structures tables, columns, indexes, and constraints.	Database Data: Manages records, business logic, and relationship behavior.
Layer	Version control for database structure.	Object-Oriented interface.
Execution	Executed via Artisan CLI commands.	Called inside application code
Key Actions	Creates tables, add columns.	Queries, creates, updates, deletes rows, and handles data casting.

3. Difference Between hasOne() and belongsTo()

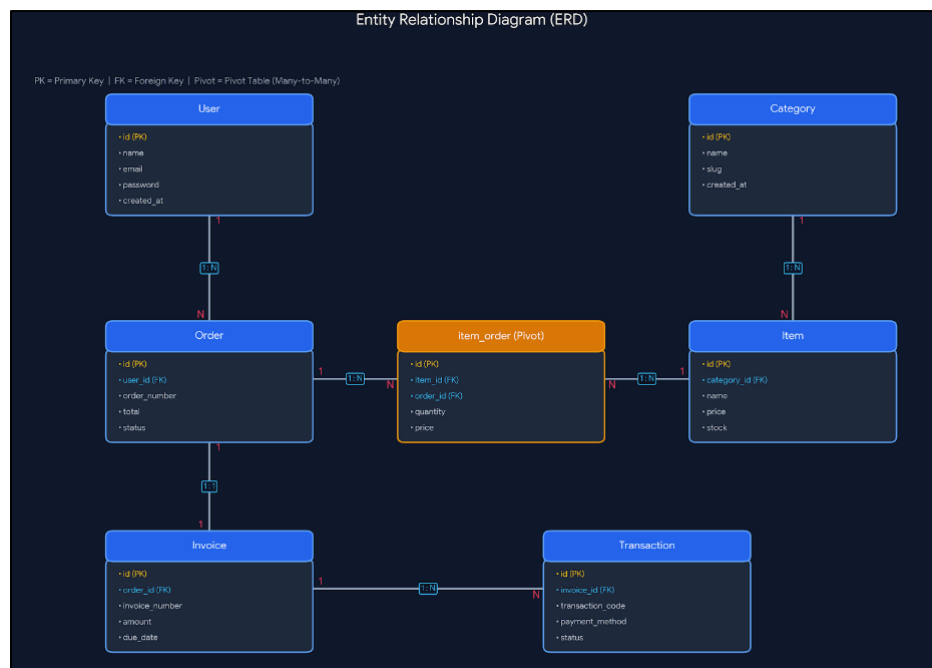
- **hasOne() (Parent/Owner Perspective):**
 - Declared on the parent model that owns the relationship.

- Indicates that another table contains a foreign key pointing back to this model.
- Example: A User hasOne(Profile::class). The profiles table stores the user_id foreign key.
- **belongsTo() (Child/Dependent Perspective):**
 - Declared on the child model that holds the foreign key column in its own table.
 - Points back to the parent record.
 - Example: A Profile belongsTo(User::class). The foreign key user_id resides inside the profiles table.

4. Difference Between up() and down() Methods in a Migration

- **up() Method:** Executed when running php artisan migrate. It applies schema changes to the database.
- **down() Method:** Executed during. It reverses the operations performed in up() (e.g., dropping tables or removing added columns) to safely undo database changes.

Parte 2:





```
user@laravel-app:~$ php artisan migrate
```

```
INFO  Running migrations.
```

2014_10_12_000000_create_users_table	DONE	12.45ms
2024_01_15_000001_create_categories_table	DONE	8.12ms
2024_01_15_000002_create_items_table	DONE	15.30ms
2024_01_15_000003_create_orders_table	DONE	11.20ms
2024_01_15_000004_create_item_order_table	DONE	9.84ms
2024_01_15_000005_create_invoices_table	DONE	10.15ms
2024_01_15_000006_create_transactions_table	DONE	14.02ms

```
user@laravel-app:~$ php artisan migrate:status
```

Ran?	Migration	Batch
Yes	2014_10_12_000000_create_users_table	[1]
Yes	2024_01_15_000001_create_categories_table	[1]
Yes	2024_01_15_000002_create_items_table	[1]
Yes	2024_01_15_000003_create_orders_table	[1]
Yes	2024_01_15_000004_create_item_order_table	[1]
Yes	2024_01_15_000005_create_invoices_table	[1]
Yes	2024_01_15_000006_create_transactions_table	[1]